Oussama Essalehi

Data Analyst

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Education

Université Toulouse III – Paul Sabatier Sept. 2024 – Aug. 2026
Master's in MIAGE – Data Engineering and Protection (IDP) *Toulouse, France*

Courses: *Data Science, Data Engineering, Machine Learning, Big Data, Gen AI, Cloud & DevOps, Agile Project Management, Data Governance*

Experience

Data Analyst Sept. 2024 – Aug. 2026
Wewise *Toulouse, France*

- Translated the needs of 4 business departments (sales, purchasing, logistics, management) into around thirty management KPIs, computed from ERP data.
- Designed 8 Power BI dashboards used every week by about forty employees, on a 15-table star schema with 60+ DAX measures.
- Cleaned and validated 100,000+ rows from the ERP and business Excel files with SQL Server queries and Python scripts (pandas, numpy), correcting ∼4% duplicates and inconsistencies.
- Cut report refresh time from 15 to 4 min and model size by 35% (DAX and Power Query optimisation), and automated 5 recurring tasks in VBA, saving about 6 hours of manual work per week.
- Trained around thirty users, documented the KPIs and set up 6 RLS security roles to manage access rights.

Projects

Batch & Streaming E-commerce Data Platform | Demo | Code
Python, Kafka, Spark, Airflow, dbt, PostgreSQL, Great Expectations, Docker

- Designed batch (API, CSV) and streaming (orders through Kafka) ingestion pipelines, orchestrated by Airflow with retries and alerts.
- Structured a Data Lake in a medallion architecture (bronze, silver, gold) on Delta Lake, then modelled a star-schema Data Warehouse with dbt.
- Set up automated data quality checks (Great Expectations, dbt tests): completeness, uniqueness and freshness, with anomaly tracking.
- Delivered data marts for analytics (Power BI) and data science (features for demand forecasting).
- Documented the pipelines, lineage and schemas with dbt docs, and containerised the whole stack with Docker Compose.

VéloScope Toulouse – Bike-Sharing Analysis and Forecasting | Demo | Code
Databricks, PySpark, Delta Lake, MLflow, scikit-learn, Streamlit, Power BI

- Continuous collection of the status of the 450 VélôToulouse stations (GBFS feed) and of the weather, through scheduled Databricks jobs.
- Delta Lake lakehouse in a medallion architecture (bronze, silver, gold) in PySpark and Spark SQL, quarantining inconsistent rows and alerting on the rejection rate.
- Rebuilt departures and arrivals from stock changes between two readings, as the APIs only publish the status of each station.
- Segmented the stations by clustering and forecast demand 24 h ahead with gradient boosting, benchmarked against a simple baseline and tracked in MLflow.
- Streamlit app (live map, forecasts, recommended rebalancing) and a Power BI dashboard in DirectQuery.

Immoscope France – French Housing Market 2021–2025 | Demo | Code
Python (pandas, SciPy), PostgreSQL, SQL, Docker, Power BI (DAX), React, Plotly

- Python pipeline downloading and cleaning 385 public DVF files (18M rows) down to 3.8M reliable sales, with automated tests.
- Star-schema model in PostgreSQL (Docker) and SQL analysis queries: window functions, medians, GROUPING SETS.
- Statistical analysis of the gaps between cities (Mann-Whitney tests, Bonferroni correction, effect sizes), with SQL and Python medians cross-checked.
- Delivered as a 4-page Power BI dashboard (DAX measures, budget simulator) and a bilingual React website with an interactive map.

Technical Skills

Programming: Python (pandas, numpy), SQL, Java, JavaScript
Data Engineering: Spark / PySpark, Kafka, Airflow, dbt, Delta Lake, Databricks, ETL / ELT, Great Expectations
Cloud & DevOps: AWS, Azure, Docker, Git, FastAPI
AI & LLM: scikit-learn, MLflow, Hugging Face, LangChain, RAG, LLM APIs, Streamlit
Databases: PostgreSQL, SQL Server, MySQL, MongoDB, Oracle
Visualisation & BI: Power BI (DAX), Power Query, Excel (VBA), Tableau
Languages: French, English

Interests

Chess, poker and strategy games · Running and fitness.